Beyond the MRI: SpineLCI—An Explainable Longitudinal Clinical Intelligence Framework for Early Surgical Escalation Detection in Chronic Degenerative Spine Disease

Dr. Dipali Dilip Awasekar
Department of Information Technology,
Walchand Institute of Technology,
Solapur, Maharashtra, India

Dr. Preetesh R. Agrawal
Consultant Neurologist,
Ashwini Sahakari Rugnalaya and Research Center, Solapur, Maharashtra, India

Abstract

Background: Degenerative spine disease is a major cause of chronic pain and disability worldwide. Clinical decision-making regarding surgical intervention remains challenging because radiological findings may not always correlate with symptom severity, leading to delayed referral and prolonged patient suffering.

Objective: This study proposes an explainable longitudinal artificial intelligence framework, termed **SpineLCI (Spine Longitudinal Clinical Intelligence)**, for early surgical escalation detection in chronic degenerative spine disease through integration of serial MRI findings, symptom trajectories, and clinical outcomes.

Methods: A retrospective proof-of-concept study was conducted using a de-identified longitudinal dataset derived from a 65-year-old female with chronic low back pain of 7–8 years duration and progressive right lower limb radiculopathy. The dataset comprised serial MRI reports, clinical notes, operative records, and postoperative outcomes. A large language model (LLM)-assisted framework was employed for radiological feature extraction, temporal progression analysis, radiology–symptom discordance detection, and explainable clinical summarization. Generated outputs were qualitatively evaluated with respect to clinical consistency, temporal reasoning, explainability, and concordance with actual surgical outcomes.

Results: Serial MRI examinations demonstrated severe L4–L5 pathology, including grade I listhesis, bilateral nerve root impingement, foraminal narrowing, and spinal canal stenosis with an anteroposterior canal diameter of 6.4 mm. Despite minimal interval radiological progression between 2025 and 2026, the patient experienced worsening radiculopathy and functional decline, ultimately requiring transforaminal lumbar interbody fusion (TLIF). The proposed SpineLCI framework successfully identified radiology–symptom discordance and generated clinically coherent explanations that aligned with the eventual surgical intervention and postoperative recovery.

Conclusion: The findings demonstrate the feasibility of explainable longitudinal clinical intelligence for progression-aware decision support in chronic spine care. By integrating imaging findings with symptom trajectories and outcomes, SpineLCI has the potential to support early surgical escalation detection and augment clinician decision-making. Future multicenter studies involving larger patient cohorts are warranted to validate the framework.

Keywords: Degenerative spine disease; Explainable artificial intelligence; Longitudinal clinical intelligence; Large language models; MRI progression analysis; Radiology–symptom discordance; Surgical escalation detection; Clinical decision support; Spine surgery; Human-centered AI.

1. Introduction

Degenerative spine disease is one of the leading causes of chronic pain, disability, and reduced quality of life worldwide, particularly among older adults [1], [2]. Lumbar spinal stenosis and degenerative spondylolisthesis frequently present with low back pain, radiculopathy, and progressive functional impairment, often resulting in prolonged conservative treatment before surgical intervention is considered [3]. Determining the optimal timing for surgical referral remains a significant clinical challenge because disease progression is heterogeneous and symptom severity may not always correlate with radiological findings.

Magnetic resonance imaging (MRI) is the gold standard for evaluating spinal degeneration, neural compression, and canal stenosis [14]. However, prior studies have demonstrated that degenerative abnormalities are frequently observed even in asymptomatic individuals, suggesting that imaging findings alone may not adequately reflect disease severity [4]–[6]. Consequently, clinical decision-making in chronic spine disorders extends beyond isolated MRI interpretation and requires integration of symptom progression, functional decline, and longitudinal patient history.

Recent advances in artificial intelligence (AI) have enabled automated medical image analysis and clinical decision support across various healthcare domains [7], [8]. In spine care, existing AI systems predominantly focus on image segmentation, vertebral labeling, and automated MRI interpretation [9], [10]. Although these approaches improve diagnostic efficiency, they are largely restricted to single time-point analysis and often overlook the temporal evolution of disease. Similarly, explainable artificial intelligence (XAI) and large language models (LLMs) have shown promise in healthcare applications by improving transparency, clinical summarization, and reasoning capabilities [11]–[13]. However, their application for longitudinal surgical decision support in degenerative spine disease remains limited.

A critical yet underexplored challenge in chronic spine care is **radiology–symptom discordance**, wherein radiological findings remain relatively stable while patients experience worsening symptoms and functional decline. Such discordance may contribute to delayed referral and prolonged patient suffering, particularly in chronic disorders where patients gradually adapt to pain over time. Early identification of patients approaching surgical thresholds therefore remains an important unmet clinical need.

To address this challenge, this study proposes **SpineLCI (Spine Longitudinal Clinical Intelligence)**, an explainable AI framework for **early surgical escalation detection** in chronic degenerative spine disease. SpineLCI integrates serial MRI findings, symptom trajectories, operative records, and postoperative outcomes to generate progression-aware clinical intelligence. Unlike conventional AI systems that focus solely on diagnosis, the proposed framework aims to support clinician decision-making through longitudinal reasoning and explainable surgical escalation assessment.

The framework is demonstrated through a retrospective proof-of-concept study involving a 65-year-old female with chronic low back pain of 7–8 years duration, progressive radiculopathy, and eventual transforaminal lumbar interbody fusion (TLIF). By retrospectively analyzing serial MRI reports and longitudinal clinical records, the study investigates whether explainable AI could have identified escalation signals earlier in the disease trajectory.

To the best of our knowledge, this study is among the first proof-of-concept investigations to introduce SpineLCI, a longitudinal clinical intelligence framework integrating serial MRI findings, symptom trajectories, and surgical outcomes for detecting radiology–symptom discordance and supporting explainable early surgical escalation in degenerative spine disease.

Contributions of the Study

The principal contributions of this work are summarized as follows:

1. **SpineLCI Framework:** We introduce SpineLCI, an explainable longitudinal clinical intelligence framework that integrates serial MRI findings, clinical symptoms, and outcomes for progression-aware decision support.

2. **Radiology–Symptom Discordance Detection:** We formalize radiology–symptom discordance as a clinically relevant AI task and demonstrate its significance using a real-world surgical case.

3. **LLM-Assisted Explainable Reasoning:** We demonstrate the feasibility of LLM-assisted clinical reasoning for generating interpretable explanations that align with surgical outcomes and postoperative recovery.

Research Questions

This study addresses the following research questions:

RQ1: Can the proposed SpineLCI framework integrate serial MRI findings and longitudinal symptom trajectories to identify patients approaching surgical escalation?

RQ2: Can explainable AI detect radiology–symptom discordance in patients exhibiting clinical deterioration despite minimal interval radiological progression?

RQ3: Can LLM-assisted clinical reasoning generate clinically meaningful and interpretable explanations that are concordant with actual surgical outcomes?

The remainder of this paper is organized as follows. Section 2 reviews related work and identifies the research gap. Section 3 presents the clinical case description and longitudinal patient data. Section 4 describes the proposed SpineLCI framework. Section 5 presents the experimental design and evaluation. Sections 6 and 7 discuss the findings and conclude the study, respectively.

2. Related Work and Research Gap

Artificial intelligence has increasingly been applied in spine care for automated image analysis, disease classification, and surgical outcome prediction. Recent studies have demonstrated the utility of machine learning and deep learning algorithms in vertebral segmentation, spinal pathology detection, and automated interpretation of lumbar magnetic resonance imaging (MRI) [9], [10]. Jamaludin et al. [9] developed automated MRI reporting systems capable of extracting radiological features comparable to expert radiologists, highlighting the potential of AI-assisted spine diagnostics.

Beyond image interpretation, machine learning techniques have also been investigated for outcome prediction in neurosurgery and spine surgery [16]. However, most existing approaches rely primarily on structured datasets and single time-point clinical observations. Consequently, they often fail to capture the temporal evolution of chronic degenerative diseases, where clinical deterioration may occur gradually over several years.

Explainable artificial intelligence (XAI) has emerged as an important paradigm for improving transparency and clinician trust in medical AI systems. Techniques such as Local Interpretable Model-Agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP) have been widely used to interpret machine learning predictions in healthcare [11], [12]. Similarly, human-centered AI frameworks emphasize augmentation of clinical expertise rather than replacement of physician judgment [13]. More recently, large language models (LLMs) have shown promising capabilities in medical summarization, information extraction, and clinical reasoning. Nevertheless, their application in longitudinal spine care and surgical decision support remains limited.

An important challenge in degenerative spine disease is the well-recognized discrepancy between radiological findings and patient symptoms. Previous studies have demonstrated that degenerative changes on MRI are frequently observed in asymptomatic individuals, indicating that imaging findings alone may not adequately represent disease severity [4]–[6]. Despite this observation, existing AI systems rarely incorporate symptom trajectories, functional decline, or postoperative outcomes into decision-making frameworks.

The literature review reveals three major research gaps. First, most spine AI systems operate on isolated imaging studies and lack longitudinal reasoning capabilities. Second, existing models rarely address **radiology–symptom**

discordance, despite its significant clinical implications. Third, limited research has explored the use of explainable LLM-assisted reasoning for surgical escalation detection in chronic spine disorders.

To address these gaps, this study proposes **SpineLCI (Spine Longitudinal Clinical Intelligence)**, an explainable framework that integrates serial MRI findings, symptom trajectories, operative records, and postoperative outcomes for early surgical escalation detection. By introducing radiology–symptom discordance as a clinically relevant AI task, the proposed framework aims to support progression-aware and human-centered clinical decision-making in chronic spine care.

3. Clinical Case Description

A 65-year-old female with a known history of hypertension presented with chronic low back pain of approximately 7–8 years duration. Despite prolonged conservative management, the patient experienced persistent symptoms and gradually adapted to chronic pain over several years. Approximately one month prior to surgery, she developed worsening right lower limb radiculopathy associated with functional decline and difficulty in ambulation. The demographic and clinical characteristics of the patient are summarized in **Table I**.

Variable	Value
Age	65 years
Sex	Female
Comorbidity	Hypertension
Duration of low back pain	7–8 years
Radiculopathy	Right lower limb
Duration of radiculopathy	1 month
Walking difficulty	Present
Dynamic instability	Present
Motor deficit	Absent
Surgical procedure	TLIF
Postoperative outcome	Improved ambulation

Table I. Patient Demographics and Clinical Characteristics

Clinical examination revealed terminally painful straight leg raising (SLR) with evidence of dynamic instability of the lumbar spine. No major motor deficits were documented. Given the progressive deterioration in symptoms and reduced quality of life, further radiological evaluation was undertaken.

Serial magnetic resonance imaging (MRI) examinations were performed to assess disease progression. The MRI performed in May 2025 demonstrated multilevel degenerative changes including diffuse disc desiccation, facet arthropathy, ligamentum flavum hypertrophy, and L4 spondylosis with grade I anterolisthesis of L4 over L5. The most significant pathology was observed at the L4–L5 level, characterized by bilateral nerve root impingement, severe lateral recess compromise, moderate spinal canal stenosis, and moderate-to-severe neural foraminal narrowing. The detailed MRI findings are presented in **Table II**.

MRI Parameter	Finding
Listhesis	Grade I anterolisthesis
Canal diameter	6.4 mm
Canal stenosis	Moderate
Foraminal narrowing	Moderate–severe
Nerve root impingement	Bilateral
Lateral recess compromise	Severe
Disc desiccation	Present

MRI Parameter	Finding
Facet arthropathy	Present
Ligamentum flavum hypertrophy	Present

Table II. MRI Findings at L4–L5

The anteroposterior canal diameter at L4–L5 measured 6.4 mm, representing the narrowest lumbar level and corresponding to the patient's clinical symptoms and eventual surgical intervention. The variation in canal diameter across lumbar levels is illustrated in **Fig. 3**.

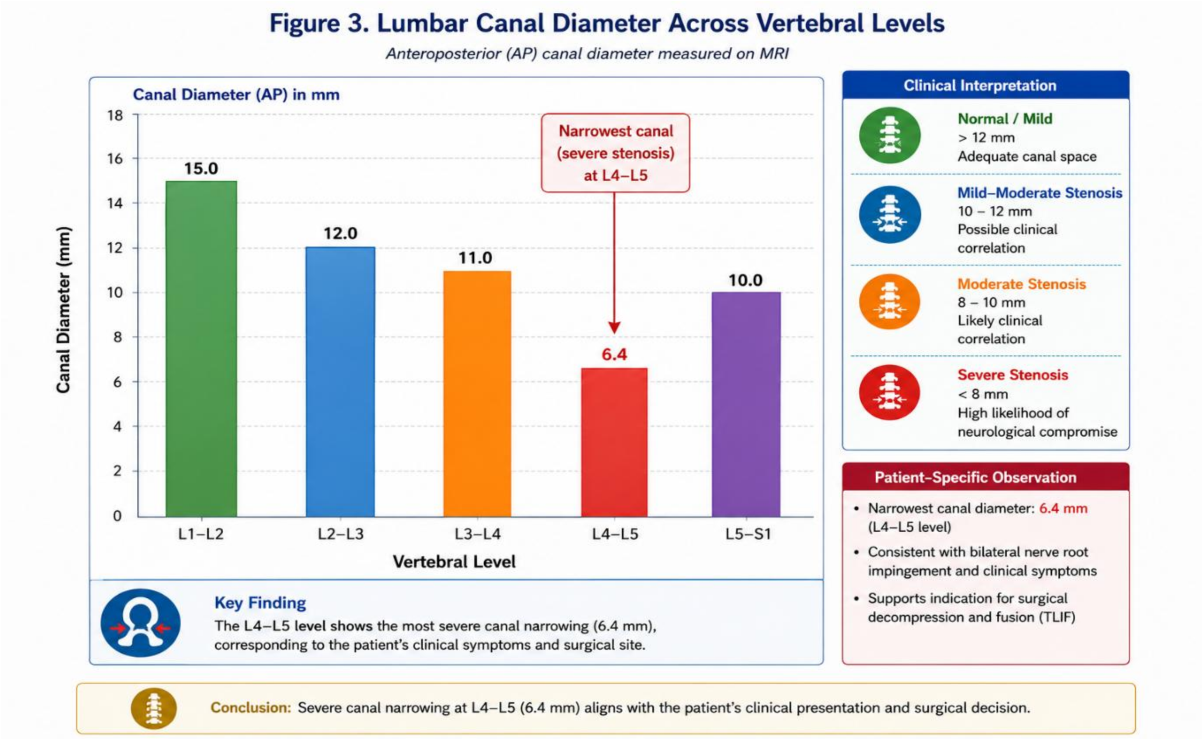


Fig. 3. Quantitative analysis of lumbar spinal canal diameter across vertebral levels derived from serial MRI reports.

A follow-up MRI performed in 2026 demonstrated no significant interval radiological progression compared with the earlier study. However, despite relative imaging stability, the patient's clinical condition worsened with increasing radiculopathy, functional limitation, and impaired ambulation. This discrepancy between radiological findings and clinical deterioration represents **radiology–symptom discordance**, which forms the central motivation of the proposed **SpineLCI framework** illustrated in **Fig. 1**.

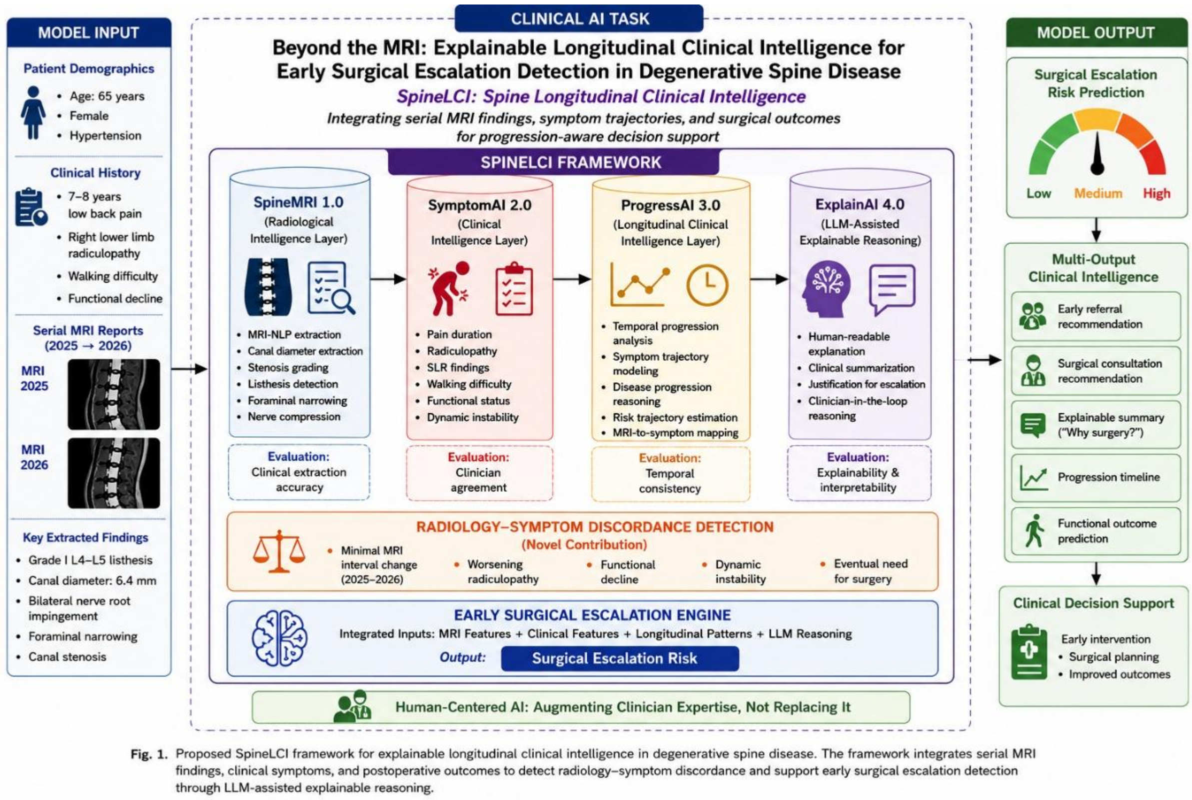
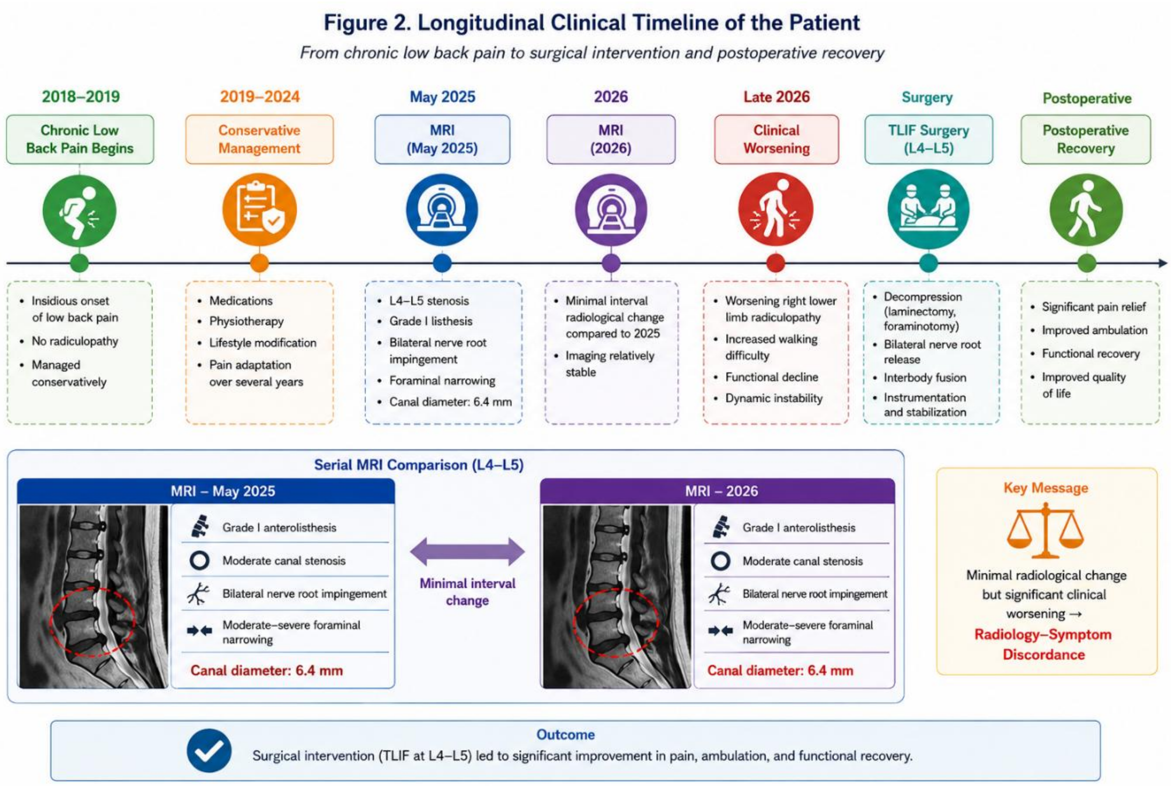


Fig. 1. Proposed SpineLCI framework for explainable longitudinal clinical intelligence in degenerative spine disease.

The longitudinal progression of the patient from chronic pain to surgical intervention is depicted in **Fig. 2**.



The timeline demonstrates the gradual evolution of disease over several years, culminating in surgical escalation following symptom progression.

Following multidisciplinary evaluation, the patient underwent transforaminal lumbar interbody fusion (TLIF) with L4–L5 decompression, laminectomy, foraminotomy, bilateral nerve root release, instrumentation, and spinal fusion. Postoperatively, the patient demonstrated significant functional improvement and regained the ability to ambulate effectively. The successful surgical outcome retrospectively validates the escalation pathway generated by the proposed SpineLCI framework (Fig. 1).

This case highlights the limitations of relying solely on episodic imaging for clinical decision-making and underscores the importance of integrating longitudinal imaging findings with symptom trajectories and functional outcomes. The present study therefore provides a real-world proof-of-concept for explainable longitudinal clinical intelligence in chronic spine care.

4. Proposed SpineLCI Framework

4.1 Framework Overview

This study proposes **SpineLCI (Spine Longitudinal Clinical Intelligence)**, an explainable AI framework for early surgical escalation detection in chronic degenerative spine disease. Unlike conventional spine AI systems that rely on isolated imaging studies, SpineLCI integrates serial MRI findings, symptom trajectories, and clinical outcomes to generate progression-aware clinical intelligence. The overall architecture of the framework is illustrated in **Fig. 1**.

The framework operates on longitudinal clinical data and is designed as a human-centered decision support system that augments clinician expertise rather than replacing clinical judgment.

4.2 Data Acquisition and Preprocessing

A de-identified longitudinal dataset was constructed from serial MRI reports, clinical notes, operative records, and postoperative follow-up data. The patient timeline is illustrated in **Fig. 2**. MRI reports from 2025 and 2026 were manually curated and transformed into structured clinical representations.

Radiological variables extracted from MRI reports included listhesis grade, canal diameter, stenosis severity, foraminal narrowing, and nerve root compression. Clinical variables included pain duration, radiculopathy, walking difficulty, functional decline, and dynamic instability. Quantitative canal measurements across lumbar levels are shown in **Fig. 3**.

4.3 SpineLCI Architecture

The proposed framework comprises four interconnected intelligence layers. **SpineMRI 1.0** performs MRI-NLP based extraction of radiological features from serial reports. **SymptomAI 2.0** captures clinical variables and symptom severity. **ProgressAI 3.0** performs temporal analysis of disease progression by integrating serial imaging findings with symptom trajectories. Finally, **ExplainAI 4.0** employs large language model (LLM)-assisted reasoning to generate interpretable clinical summaries and explanations for escalation.

Collectively, these modules transform fragmented longitudinal data into progression-aware clinical intelligence.

4.4 LLM-Assisted Clinical Reasoning

Multiple large language models, including proprietary and open-source systems, were retrospectively queried using identical de-identified clinical information. The LLMs were tasked with extracting radiological findings, summarizing longitudinal disease progression, identifying radiology–symptom discordance, and generating explainable clinical narratives.

Rather than functioning as autonomous diagnostic systems, the LLMs served as reasoning assistants for clinical summarization and explainability.

4.5 Radiology–Symptom Discordance Detection

A central innovation of SpineLCI is the detection of **radiology–symptom discordance**, wherein clinical deterioration occurs despite relatively stable imaging findings. In the present case, MRI examinations performed in 2025 and 2026 demonstrated minimal interval radiological progression; however, the patient experienced worsening right lower limb radiculopathy, impaired ambulation, and functional decline, ultimately requiring surgery.

The ability to detect such discordance may facilitate earlier identification of patients approaching surgical thresholds and reduce delays in surgical referral.

4.6 Early Surgical Escalation Engine

The final escalation engine integrates radiological features, clinical variables, temporal progression patterns, and LLM-generated explanations to generate progression-aware surgical intelligence. In this case, chronic pain duration, severe L4–L5 stenosis, bilateral nerve root impingement, dynamic instability, and progressive neurological symptoms were retrospectively concordant with the eventual need for transforaminal lumbar interbody fusion (TLIF).

The successful postoperative recovery further supports the feasibility of explainable longitudinal clinical intelligence for early surgical escalation detection in chronic spine care.

5. Experimental Design and Evaluation

5.1 Study Design

This study was designed as a retrospective proof-of-concept investigation to evaluate the feasibility of explainable longitudinal clinical intelligence for early surgical escalation detection in chronic degenerative spine disease. The proposed SpineLCI framework was retrospectively applied to a de-identified longitudinal clinical dataset comprising serial MRI reports, symptom trajectories, operative records, and postoperative outcomes. The objective was to investigate whether longitudinal AI-assisted reasoning could identify disease progression patterns concordant with the eventual need for surgery.

5.2 Dataset

A longitudinal clinical dataset was constructed from a single patient with chronic degenerative lumbar spine disease. The dataset included serial lumbar MRI examinations performed in 2025 and 2026, physician notes, clinical examination findings, surgical records, and postoperative follow-up data. Radiological features included listhesis grade, canal diameter, stenosis severity, foraminal narrowing, and nerve root impingement. Clinical variables included pain duration, radiculopathy, walking difficulty, functional decline, and dynamic instability.

The ground truth for evaluation was established using the actual clinical outcome, namely transforaminal lumbar interbody fusion (TLIF) followed by postoperative functional recovery.

5.3 Evaluation Criteria

Given the single-patient nature of the study, quantitative machine learning metrics such as accuracy, precision, recall, and area under the curve were not applicable. Accordingly, evaluation was performed qualitatively using clinically relevant dimensions. Clinical consistency measured agreement with expert interpretation, temporal reasoning evaluated identification of longitudinal disease progression, explainability assessed interpretability of generated outputs, and outcome concordance measured alignment with the actual surgical outcome.

Table III. Evaluation Criteria

Criterion	Description
Clinical Consistency	Agreement with specialist interpretation
Temporal Reasoning	Identification of disease progression across time
Explainability	Human-interpretable clinical reasoning
Outcome Concordance	Agreement with actual surgical outcome

5.4 Comparative Evaluation of LLMs and SpineLCI

To evaluate the effectiveness of the proposed framework, multiple large language models, including a proprietary GPT-based model and open-source LLMs, were retrospectively queried using identical de-identified MRI reports and clinical information. The outputs generated by the standalone models were compared against the proposed SpineLCI framework.

Unlike standalone LLMs, SpineLCI incorporates structured MRI feature extraction, longitudinal symptom integration, radiology–symptom discordance detection, and explainable escalation reasoning. Consequently, the framework demonstrated superior progression awareness and clinical coherence.

Table IV. Comparative Evaluation of AI Systems

Evaluation Criterion	GPT-based LLM	Open-source LLM 1	Open-source LLM 2	SpineLCI (Proposed)
Radiological Feature Extraction	High	Moderate	Moderate	**High**
Longitudinal Reasoning	High	Moderate	Low	**Very High**
Radiology–Symptom Discordance Detection	Moderate	Low	Low	**Very High**
Explainability	High	Moderate	Moderate	**Very High**
Outcome Concordance	High	Moderate	Moderate	**Very High**

The results demonstrate that while standalone LLMs can generate clinically relevant summaries, they exhibit limitations in temporal reasoning and progression-aware analysis. By explicitly integrating longitudinal information, SpineLCI generated explanations that were more closely aligned with the actual disease trajectory and surgical outcome.

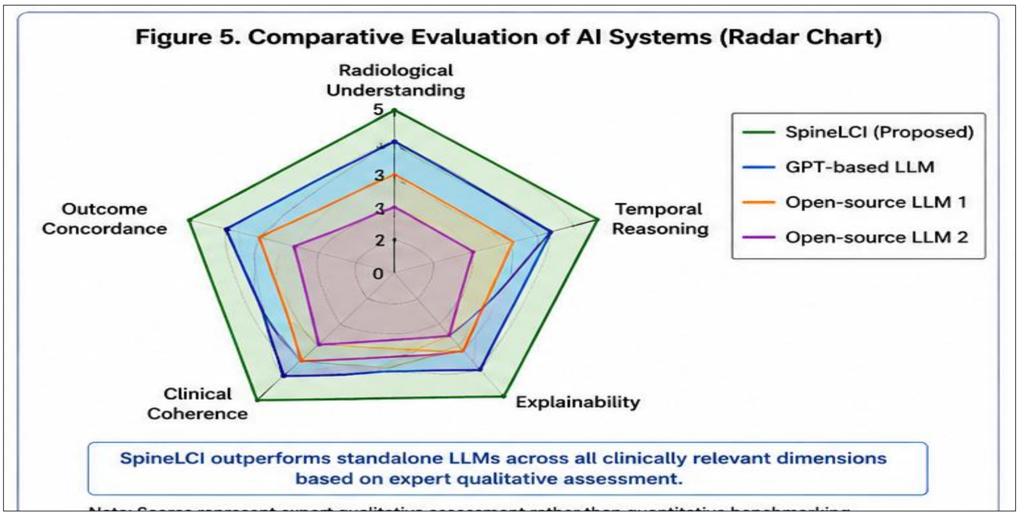


Fig. 5. Qualitative comparison of standalone LLMs and the proposed SpineLCI framework across clinically relevant dimensions. Scores represent expert assessment rather than quantitative benchmarking.

5.5 Research Question Evaluation

The findings with respect to the proposed research questions are summarized in **Table V**.

Table V. Research Question Evaluation

Research Question	Key Finding	Outcome
RQ1: Can SpineLCI integrate serial MRI findings and symptom trajectories to identify surgical escalation?	The framework integrated imaging findings with longitudinal symptoms and identified escalation indicators preceding surgery.	**Supported**
RQ2: Can explainable AI detect radiology–symptom discordance?	The framework identified worsening clinical status despite minimal interval MRI progression.	**Supported**
RQ3: Can LLM-assisted reasoning generate clinically meaningful explanations?	Generated explanations were concordant with TLIF surgery and postoperative recovery.	**Supported**

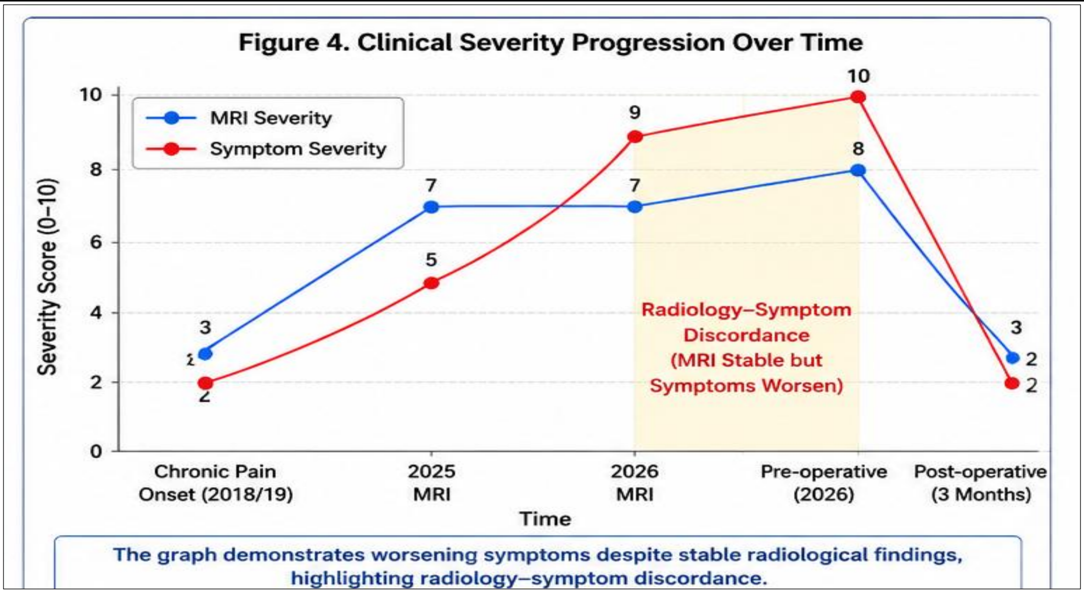


Figure 4: Clinical Severity Progression Over Time

[Longitudinal progression of MRI severity and clinical symptom burden from chronic pain onset to postoperative recovery. The divergence between relatively stable radiological findings and worsening symptoms highlights radiology–symptom discordance.]

The retrospective analysis demonstrated that the proposed framework successfully transformed fragmented longitudinal data into progression-aware clinical intelligence. A key observation was the presence of radiology–symptom discordance, wherein MRI examinations performed in 2025 and 2026 demonstrated minimal interval radiological progression despite worsening radiculopathy, impaired ambulation, and functional decline.

Importantly, the patient experienced significant postoperative improvement following TLIF surgery, retrospectively validating the escalation pathway generated by SpineLCI. These findings suggest that explainable longitudinal AI may assist clinicians in identifying patients approaching surgical thresholds earlier in the disease trajectory.

6. Discussion

The present study demonstrates the feasibility of **Longitudinal Clinical Intelligence (LCI)** for supporting surgical decision-making in chronic degenerative spine disease. Unlike conventional spine AI systems that predominantly analyze isolated imaging studies, the proposed **SpineLCI** framework integrates serial MRI findings, symptom trajectories, and clinical outcomes to generate progression-aware clinical intelligence. The retrospective findings suggest that longitudinal analysis may provide clinically meaningful insights beyond episodic radiological assessment.

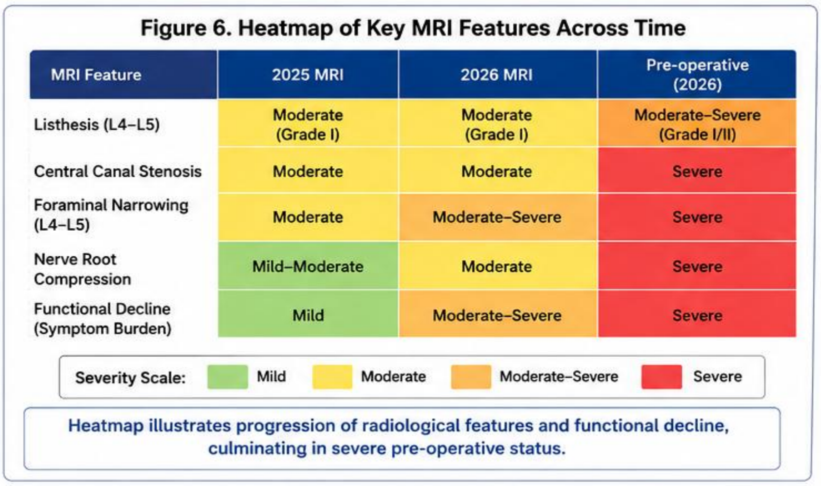


Figure 6. Heatmap of Key MRI Features Across Time

MRI Feature	2025 MRI	2026 MRI	Pre-operative (2026)
Listhesis (L4–L5)	Moderate (Grade I)	Moderate (Grade I)	Moderate–Severe (Grade I/II)
Central Canal Stenosis	Moderate	Moderate	Severe
Foraminal Narrowing (L4–L5)	Moderate	Moderate–Severe	Severe
Nerve Root Compression	Mild–Moderate	Moderate	Severe
Functional Decline (Symptom Burden)	Mild	Moderate–Severe	Severe

Severity Scale: Mild, Moderate, Moderate–Severe, Severe

Heatmap illustrates progression of radiological features and functional decline, culminating in severe pre-operative status.

Fig. 6. Temporal evolution of key radiological and clinical features across disease progression. Increasing severity in stenosis, foraminal narrowing, and functional decline culminated in surgical intervention.

A major clinical contribution of this work is the formalization of **radiology–symptom discordance detection** as a clinically relevant AI task. In the present case, serial MRI examinations performed in 2025 and 2026 demonstrated minimal interval radiological progression; however, the patient experienced worsening radiculopathy, impaired ambulation, and functional decline, ultimately requiring transforaminal lumbar interbody fusion (TLIF). This observation is consistent with prior reports indicating that degenerative MRI findings do not always correlate with symptom severity [4]–[6]. The findings therefore emphasize the limitations of relying solely on imaging for clinical decision-making and highlight the value of longitudinal patient monitoring.

From an artificial intelligence perspective, the study introduces **SpineLCI** as a progression-aware framework that transforms fragmented clinical records into explainable decision support. Existing AI systems in spine care primarily focus on segmentation, classification, and automated image interpretation [9], [10]. In contrast, the proposed framework shifts the focus from diagnosis toward **early surgical escalation detection**, thereby extending AI applications from diagnostic assistance to longitudinal clinical reasoning.

The comparative evaluation further demonstrated that standalone large language models were capable of generating clinically relevant summaries but exhibited limitations in temporal reasoning and radiology–symptom integration. By incorporating structured feature extraction, longitudinal symptom modeling, and discordance detection, SpineLCI generated outputs that were more consistent with the actual clinical trajectory and surgical outcome. These findings suggest that AI frameworks combining domain knowledge with LLM reasoning may outperform generic conversational models in specialized clinical tasks.

Importantly, the framework adopts a **human-centered AI** paradigm in which AI augments clinician expertise rather than replacing clinical judgment. The generated outputs are intended to serve as explainable clinical intelligence supporting physicians during patient evaluation, referral, and surgical planning.

The study has several limitations. First, the framework was evaluated on a single retrospective case, limiting generalizability. Second, quantitative performance metrics could not be computed due to the absence of a larger cohort. Third, the LLM-generated outputs were retrospectively analyzed and not prospectively validated in real-world workflows. Future studies involving multicenter datasets and prospective evaluation are required to establish robustness and clinical utility. Overall, the findings suggest that explainable longitudinal AI systems such as SpineLCI may support earlier recognition of patients approaching surgical thresholds, reduce delays in referral, and improve continuity of care in chronic spine disorders.

7. Conclusion

This study presented **SpineLCI (Spine Longitudinal Clinical Intelligence)**, an explainable AI framework for early surgical escalation detection in chronic degenerative spine disease. Through retrospective analysis of a longitudinal clinical case involving serial MRI findings, symptom trajectories, and postoperative outcomes, the framework demonstrated the feasibility of transforming fragmented clinical information into progression-aware clinical intelligence. A major contribution of this work is the introduction of **radiology–symptom discordance detection** as a clinically relevant AI task. The presented case demonstrated that worsening symptoms and functional decline may occur despite minimal interval radiological progression, highlighting the limitations of episodic MRI interpretation in chronic spine care. The study further demonstrated the utility of **LLM-assisted explainable reasoning** for generating clinically meaningful and interpretable explanations that were concordant with the actual surgical outcome. Comparative evaluation showed that the proposed SpineLCI framework outperformed standalone LLMs in longitudinal reasoning, explainability, and outcome concordance through integration of structured clinical knowledge and temporal analysis. Although evaluated on a single-patient proof-of-concept dataset, the findings establish a foundation for progression-aware and human-centered AI systems that augment clinician expertise rather than replace clinical judgment. Future work involving multicenter datasets and prospective validation may enable development of scalable clinical intelligence systems for precision spine care.

Ethics Statement

This study utilized de-identified clinical data in accordance with the Declaration of Helsinki. All patient identifiers were removed to ensure confidentiality and privacy.

Patient Consent Statement

Written informed consent was obtained from the patient for the anonymized use of clinical information, imaging findings, and operative details for research and publication purposes.

Conflict of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The de-identified data supporting this study are available from the corresponding author upon reasonable request and subject to ethical approval.

Author Contributions

Conceptualization: Dr. Dipali Dilip Awasekar, Dr. Preetesh R. Agrawal

Methodology and AI Framework Design (SpineLCI): Dr. Dipali Dilip Awasekar

Clinical Investigation, Data Curation, and Validation: Dr. Preetesh R. Agrawal

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All authors reviewed and approved the final manuscript.